Hermite Polynomials

The Hermite polynomials are defined via Rodrigues' formula:

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} \tag{1}$$

and they satisfy the differential equation

$$\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2ny = 0 \tag{2}$$

which appears in the context of solving for the energy eigenstates of the quantum harmonic oscillator. The energy eigenstates of the quantum harmonic oscillator are given by

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right) \frac{1}{\sqrt{2^n n!}} H_n\left(\sqrt{\frac{m\omega}{\hbar}}\, x\right) e^{-m\omega x^2/2\hbar}$$

Properties

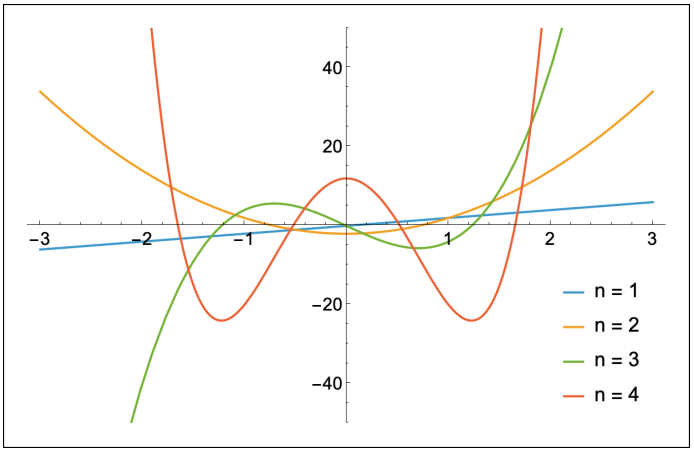
(a) Orthogonality: $\displaystyle\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx = 0 \quad (m \neq n)$

(b) Recurrence relation: $H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$

(c) Derivative property: $\dfrac{dH_n(x)}{dx} = 2nH_{n-1}(x)$

(d) Generating function: $\exp\left(2tx - x^2\right) = \displaystyle\sum_{n=0}^{\infty} \frac{t^n}{n!} H_n(x)$

First Few Hermite Polynomials



$$H_0(x) = 1$$
$$H_1(x) = 2x$$
$$H_2(x) = 4x^2 - 2$$
$$H_3(x) = 8x^3 - 12x$$
$$H_4(x) = 16x^4 - 48x^2 + 12$$
$$H_5(x) = 32x^5 - 160x^3 + 120x$$

Legendre Polynomials

Legendre polynomials $P_\ell(x)$ are given by another Rodrigues formula:

$$P_\ell(x) = \frac{1}{2^\ell \ell!} \frac{d^\ell}{dx^\ell}(x^2-1)^\ell \tag{3}$$

and they satisfy the differential equation

$$\frac{d}{dx}\left[(1-x^2)\frac{dP_\ell}{dx}\right] + \ell(\ell+1)P_\ell = 0 \tag{4}$$

which arises in the context of solving for the eigenstates of angular momentum when solutions exhibit azimuthal symmetry.

Properties

(a) Definite parity: $P_\ell(-x) = (-1)^\ell P_\ell(x)$

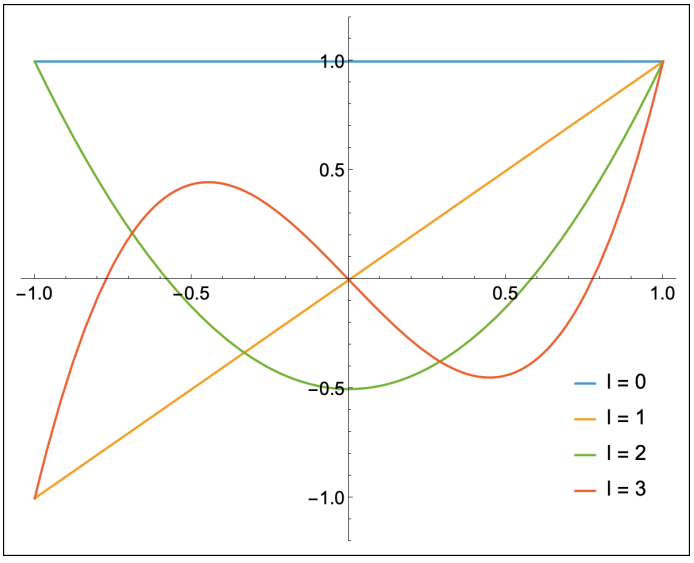
(a) Orthogonality: $\displaystyle\int_{-1}^{1} P_\ell(x)P'_\ell(x)dx = \int_0^\pi P_\ell(\cos\theta)P'_\ell(\cos\theta)\sin\theta d\theta = \frac{2}{2\ell+1}\delta_{\ell\ell'}$

(b) Recurrence relation: $(\ell+1)P_{\ell+1}(x) = (2\ell+1)xP_\ell(x) - \ell P_{\ell-1}(x)$

(c) Derivative property: $\displaystyle(1-x^2)\frac{dP_\ell(x)}{dx} = \ell\left(P_{\ell-1}(x) - xP_\ell(x)\right)$

(d) Generating function: $\displaystyle\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{\ell=0}^{\infty} P_\ell(x)t^\ell, \quad |t|<1, x\in[-1,1]$

First Few Legendre Polynomials



$$P_0(x) = 1$$
$$P_1(x) = x$$
$$P_2(x) = \frac{1}{2}\left(3x^2-1\right)$$
$$P_3(x) = \frac{1}{2}\left(5x^3-3x\right)$$
$$P_4(x) = \frac{1}{8}\left(35x^4-30x^2+3\right)$$
$$P_5(x) = \frac{1}{8}\left(63x^5-70x^3+15x\right)$$

Associated Legendre Polynomials

The associated Legendre polynomials are defined as

$$P_\ell^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x), \quad m \geq 0 \tag{5}$$

and they satisfy the associated Legendre equation

$$\frac{d}{dx}\left[(1-x^2)\frac{dP_\ell^m(x)}{dx}\right] + \left[\ell(\ell+1) - \frac{m^2}{1-x^2}\right] P_\ell^m(x) = 0 \tag{6}$$

which arises in the context of solving for the eigenstates of angular momentum. More precisely, they are the acceptable solutions to the angular part of the angular momentum eigenvalue problem in spherical coordinates.

Properties

(a) Negative m states: $P_\ell^{-m}(x) = (-1)^m \dfrac{(\ell-m)!}{(\ell+m)!} P_\ell^m(x)$

(b) Parity: $P_\ell^m(-x) = (-1)^{\ell-m} P_\ell^m(x)$

(c) Orthogonality: $\displaystyle\int_{-1}^{1} P_\ell^m(x)\, P_{\ell'}^m(x)\, dx = \frac{2(\ell+m)!}{(2\ell+1)(\ell-m)!}\, \delta_{\ell\ell'}$

(d) Recurrence relation: $(\ell-m+1)\, P_{\ell+1}^m(x) = (2\ell+1)\, x\, P_\ell^m(x) - (\ell+m)\, P_{\ell-1}^m(x)$

(e) Derivative property: $(1-x^2)\, \dfrac{d}{dx} P_\ell^m(x) = \ell x\, P_\ell^m(x) - (\ell+m)\, P_{\ell-1}^m(x)$

First Few Associated Legendre Polynomials

$$P_\ell^m(x) = P_\ell^m(\cos\theta)$$

$P_0^0 = 1$	$P_2^0 = \frac{1}{2}(3\cos^2\theta - 1)$
$P_1^1 = -\sin\theta$	$P_3^3 = -15\sin\theta(1-\cos^2\theta)$
$P_1^0 = \cos\theta$	$P_3^2 = 15\sin^2\theta\cos\theta$
$P_2^2 = 3\sin^2\theta$	$P_3^1 = -\frac{3}{2}\sin\theta(5\cos^2\theta - 1)$
$P_2^1 = -3\sin\theta\cos\theta$	$P_3^0 = \frac{1}{2}(5\cos^3\theta - 3\cos\theta)$

Spherical Harmonics

The spherical harmonics are defined as

$$Y_\ell^m(\theta,\phi) = \sqrt{\frac{(2\ell+1)}{4\pi}\frac{(\ell-m)!}{(\ell+m)!}}\, e^{im\phi} P_\ell^m(\cos\theta) \tag{7}$$

where P_ℓ^m denotes the associated Legendre polynomials. The spherical harmonic functions are the position-space representations of the simultaneous eigenstates of $\mathbf{L}^2$ and L_z, expressed in spherical coordinates, i.e., $Y_\ell^m(\theta,\phi) = \langle\theta,\phi|\ell m\rangle$. In other words,

$$\begin{aligned}\mathbf{L}^2 Y_\ell^m(\theta,\phi) &= \ell(\ell+1)\hbar^2 Y_\ell^m(\theta,\phi)\\ L_z Y_\ell^m(\theta,\phi) &= m\hbar Y_\ell^m(\theta,\phi)\end{aligned}$$

with $\ell = 0, 1, 2, \ldots$ and $m = -\ell, -\ell+1, \ldots, \ell-1, \ell$.

Properties

(a) Orthogonality over unit sphere: $\int (Y_\ell^m)^*(Y_{\ell'}^{m'})d\Omega = \delta_{\ell\ell'}\delta_{mm'}$

(b) Completeness: $f(\theta,\phi) = \sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} c_{\ell m} Y_\ell^m(\theta,\phi)$

(c) Conjugation: $Y_\ell^{-m} = (-1)^m (Y_\ell^m)^*$

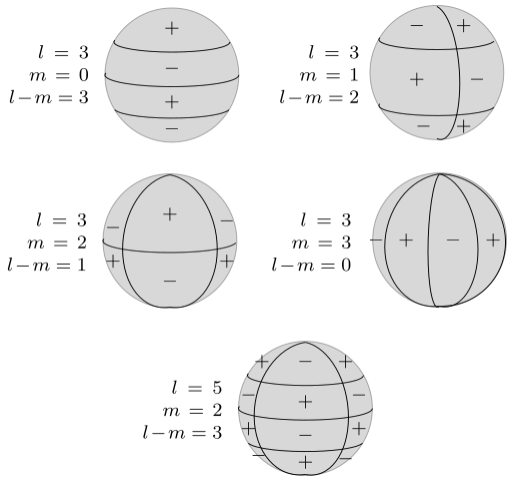
(d) Parity under spatial inversion: $Y_\ell^m(\pi-\theta,\phi+\pi) = (-1)^\ell Y_\ell^m(\theta,\phi)$

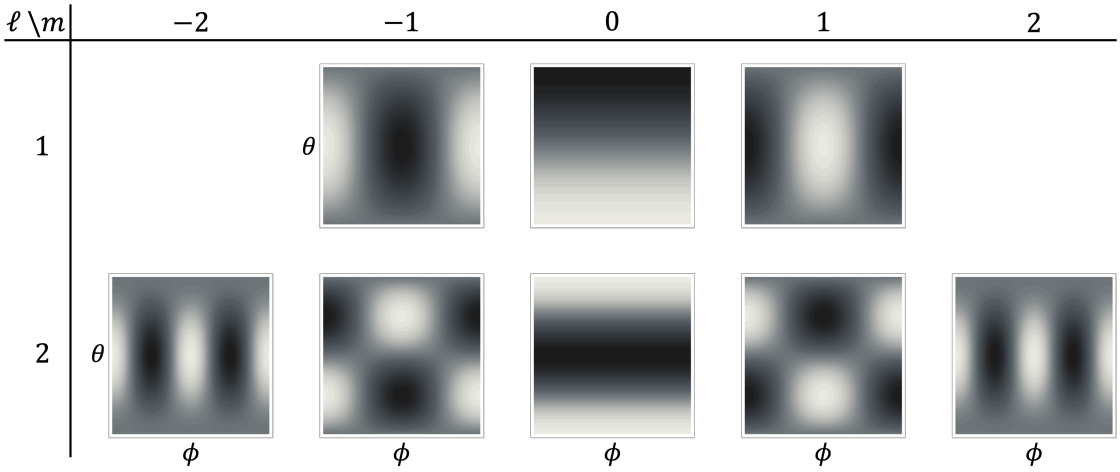
First Few Spherical Harmonics

ℓ	m	Y_ℓ^m
0	0	$\sqrt{\frac{1}{4\pi}}$
1	0	$\sqrt{\frac{3}{4\pi}}\cos\theta$
	±1	$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{\pm i\phi}$
2	0	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta - 1)$
	±1	$\mp\sqrt{\frac{15}{8\pi}}\sin\theta\cos\theta\, e^{\pm i\phi}$
	±2	$\sqrt{\frac{15}{32\pi}}\sin^2\theta\, e^{\pm 2i\phi}$

Visualizing Spherical Harmonics

Spherical harmonics are related to standing waves on the unit sphere, and we can visualize them by considering their nodal lines, i.e., the sets of all points where $\mathrm{Re}[Y_\ell^m] = 0$. The nodal lines are composed of ℓ circles; there are $|m|$ circles along longitudes and $\ell - |m|$ circles along latitudes. Here's a sketch of $\mathrm{Re}[Y_\ell^m]$ on the unit sphere and its nodal lines:





Density plots of $\mathrm{Re}[Y_\ell^m(\theta, \phi)]$ for various values of ℓ and m. Each small panel is a map of the spherical harmonic over angular coordinates, where the horizontal axis represents the azimuthal angle ϕ, running from 0 to 2π and describing rotation around the vertical (z) axis, while the vertical axis represents the polar angle θ, running from 0 at the north pole to π at the south pole; together, these axes parameterize the surface of the sphere, so the plot is effectively an "unwrapped" view of the sphere with ϕ controlling left–right variation around the equator and θ controlling top–bottom variation from pole to pole.

Laguerre Polynomials

The (ordinary) Laguerre polynomials are defined as

$$L_q(x) = \frac{e^x}{q!}\frac{d^q}{dx^q}\left(e^{-x}x^q\right) \tag{8}$$

and they satisfy the differential equation

$$x\frac{d^2y}{dx^2} + (1-x)\frac{dy}{dx} + ny = 0 \tag{9}$$

while the associated Laguerre polynomials, denoted $L_q^p(x)$, satisfy the differential equation

$$x\frac{d^2y}{dx^2} + (p+1-x)\frac{dy}{dx} + ny = 0, \qquad \text{where } p > -1 \tag{10}$$

The associated Laguerre polynomials can be defined explicitly as

$$L_q^p(x) = \frac{x^{-p}e^x}{q!}\frac{d^q}{dx^q}\left(e^{-x}x^{q+p}\right) \tag{11}$$

These polynomials arise in problems with radial symmetry, most notably in the solution to the radial equation for the hydrogen atom.

Properties

(a) Orthogonality: $\displaystyle\int_0^\infty e^{-x}x^p L_q^p(x)L_{q'}^p(x)dx = \frac{\Gamma(q+p+1)}{q!}\delta_{qq'}$

(b) Recurrence relation: $(n+1)L_{q+1}^p(x) = (2q+p+1-x)L_q^p(x) - (q+p)L_{q-1}^p(x)$

(c) Differentiation identity: $\displaystyle\frac{d}{dx}L_q^p(x) = -L_{q-1}^{p+1}(x)$

(d) Generating function: $\displaystyle\sum_{q=0}^{\infty} L_q^p(x)\,t^q = \frac{1}{(1-t)^{p+1}}\exp\left(-\frac{xt}{1-t}\right), \quad |t|<1$

First Few Associated Laguerre Polynomials

For associated Laguerre polynomial $L_q^p(x)$:

$p\backslash q$	0	1	2	3
1	$1-x$	$2-x$	$3-x$	$4-x$
2	$1-2x+\frac{x^2}{2}$	$3-3x+\frac{x^2}{2}$	$6-4x+\frac{x^2}{2}$	$10-5x+\frac{x^2}{2}$